AETHERIA FLIGHT DYNAMICS

Falcon-9X Operations Manual & Mission Parameters (Page 1)

1. MISSION OVERVIEW
Mission Name: Orbital Survey Aurora-7
Flight Commander: Commander Sarah Jenkins
Mission Specialist: Dr. Alan Turing-Rao

2. PROPULSION SPECIFICATIONS
Propulsion System: Dual MethaLOX Raptor-4 rocket engines
Total Vacuum Thrust: 2.3 meganewtons
Propellant Mass: 420,000 kilograms of cryogenic liquid methane and LOX

3. LAUNCH WINDOW & PAYLOAD
Target Launch Date: October 30, 2026 at 06:45 UTC
Launch Complex: Pad 39-B, Space Coast Spaceport
Primary Scientific Payload: Hyperspectral Atmospheric Imager
Payload Mass: Exactly 1,420 kilograms
Primary Target Orbit: Low Earth Orbit (LEO) at 450 km inclination

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Emergency Protocols & Life Support Subsystems (Page 2)

4. LIFE SUPPORT SUBSYSTEMS
Primary Cabin Oxygen Reserve: 180 hours of nominal cabin atmosphere
Auxiliary Nitrogen Regulator: Valve N-42 located in Compartment Charlie
Nominal Cabin Pressure: 101.3 kPa (sea level equivalent)
Water Recycling Efficiency: 94.8% closed-loop reclamation

5. EMERGENCY ABORT PROCEDURES
Emergency Distress Transponder Frequency: 406.025 MHz
Distress Audio Beacon Code: AURORA-MAYDAY-7
Abort Contingency Safe Orbit Altitude: Exactly 220 kilometers above sea level
Ground Recovery Command Center: Call sign 'Blue Horizon' on secure satellite link Alpha-9